

Validation report: PD scorecard on Lending Club 36-month loans

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Summary

Out-of-time (2014–2015) AUC 0.639, Gini 0.277, KS 0.199; Hosmer–Lemeshow p-value 0; score PSI 0.003.

Findings and tables

Discrimination scorecard

sample	n	default_rate	auc	gini	ks
train	140225	0.126	0.651	0.301	0.220
holdout	35201	0.126	0.652	0.304	0.226
oot	445596	0.145	0.639	0.277	0.199

Discrimination challenger

sample	n	default_rate	auc	gini	ks
train	140225	0.126	0.705	0.410	0.295
holdout	35201	0.126	0.675	0.351	0.262
oot	445596	0.145	0.661	0.322	0.232

Calibration deciles oot

group	n	defaults	observed_dr	mean_pd	binom_pvalue
1	44560	2048	0.046	0.045	0.271
2	44560	3544	0.080	0.069	0.000
3	44559	4445	0.100	0.085	0.000
4	44560	5262	0.118	0.099	0.000
5	44559	5967	0.134	0.113	0.000
6	44560	6780	0.152	0.129	0.000
7	44559	7407	0.166	0.145	0.000
8	44560	8225	0.185	0.165	0.000
9	44559	9476	0.213	0.191	0.000
10	44560	11293	0.253	0.246	0.000

Stability

variable	psi	status
loan_amnt	0.019	stable
annual_inc	0.009	stable
dti	0.066	stable
emp_length_years	0.023	stable
fico_mid	0.058	stable
inq_last_6mths	0.034	stable
delinq_2yrs	0.026	stable
pub_rec	0.001	stable
open_acc	0.034	stable
total_acc	0.014	stable
revol_bal	0.009	stable
revol_util	0.031	stable
credit_history_years	0.054	stable
home_ownership	0.015	stable
verification_status	0.165	monitor
purpose	0.106	monitor
score	0.003	stable

El backtest

vintage	n	exposure	expected_loss	realized_loss	ratio
2014	162570	2046040750.000	216201182.388	136945537.923	1.579
2015	283026	3624300800.000	370240162.773	274011577.241	1.351

Sensitivity

scenario	expected_loss
base	586441345.161
PD x1.2	703729614.193
LGD +10pp	651504473.061
PD x1.2 & LGD +10pp	781805367.673

Coefficients

feature	coef
fico_mid	-0.797
annual_inc	-0.825
revol_util	-0.351
dti	-0.583
purpose	-1.055
inq_last_6mths	-1.069
home_ownership	-0.419

Figures

